

Towards Super-Resolution of Thermal Infrared Images to Assess Temperature Spatial Heterogeneity in Rivers.

Master's Thesis

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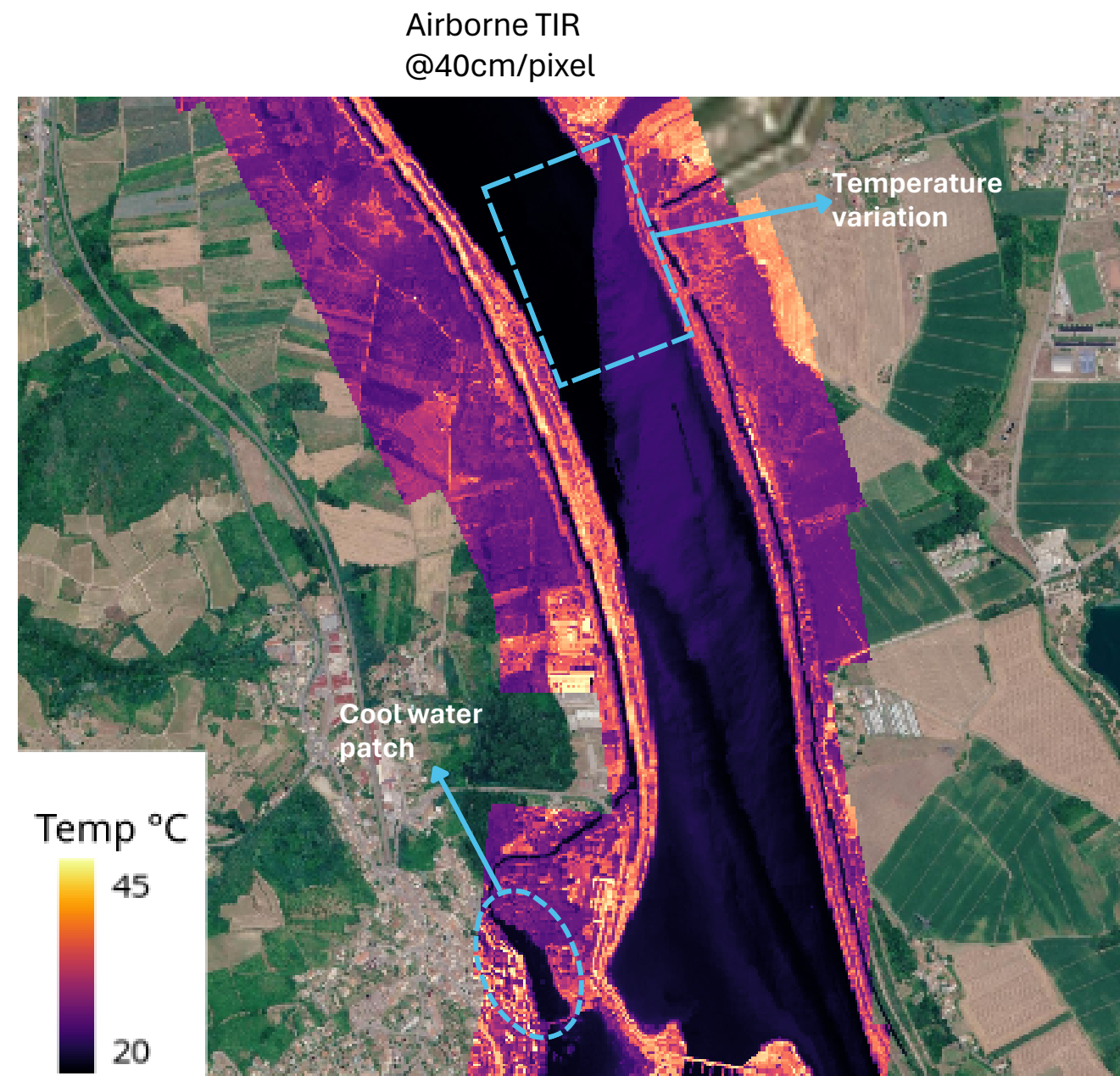
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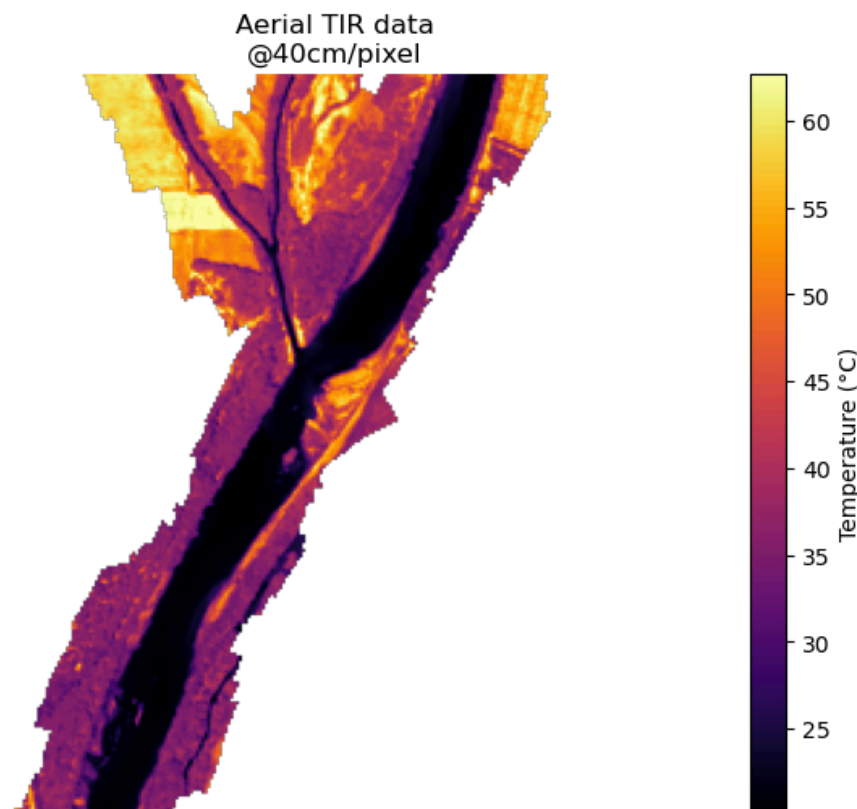
Background & Motivation



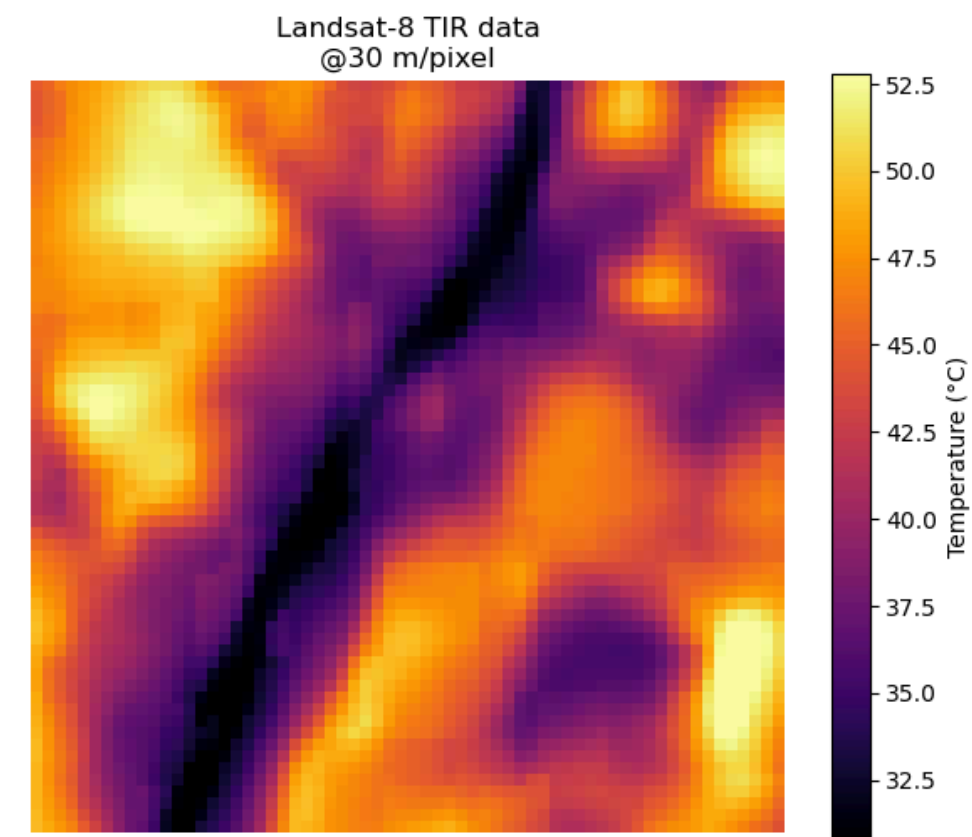
- River temperature drives biodiversity.
- It varies longitudinally providing thermal habitats (refugia) which protect temperature-sensitive species from thermal stress.
- Essential to identify the fine-scale thermal habitats especially in the context of warming climate.

Background & Motivation

Challenge: Spatial and temporal resolution trade-off



- Aerial imagery: high-resolution but expensive.
- Covers small spatial extent with limited temporal coverage



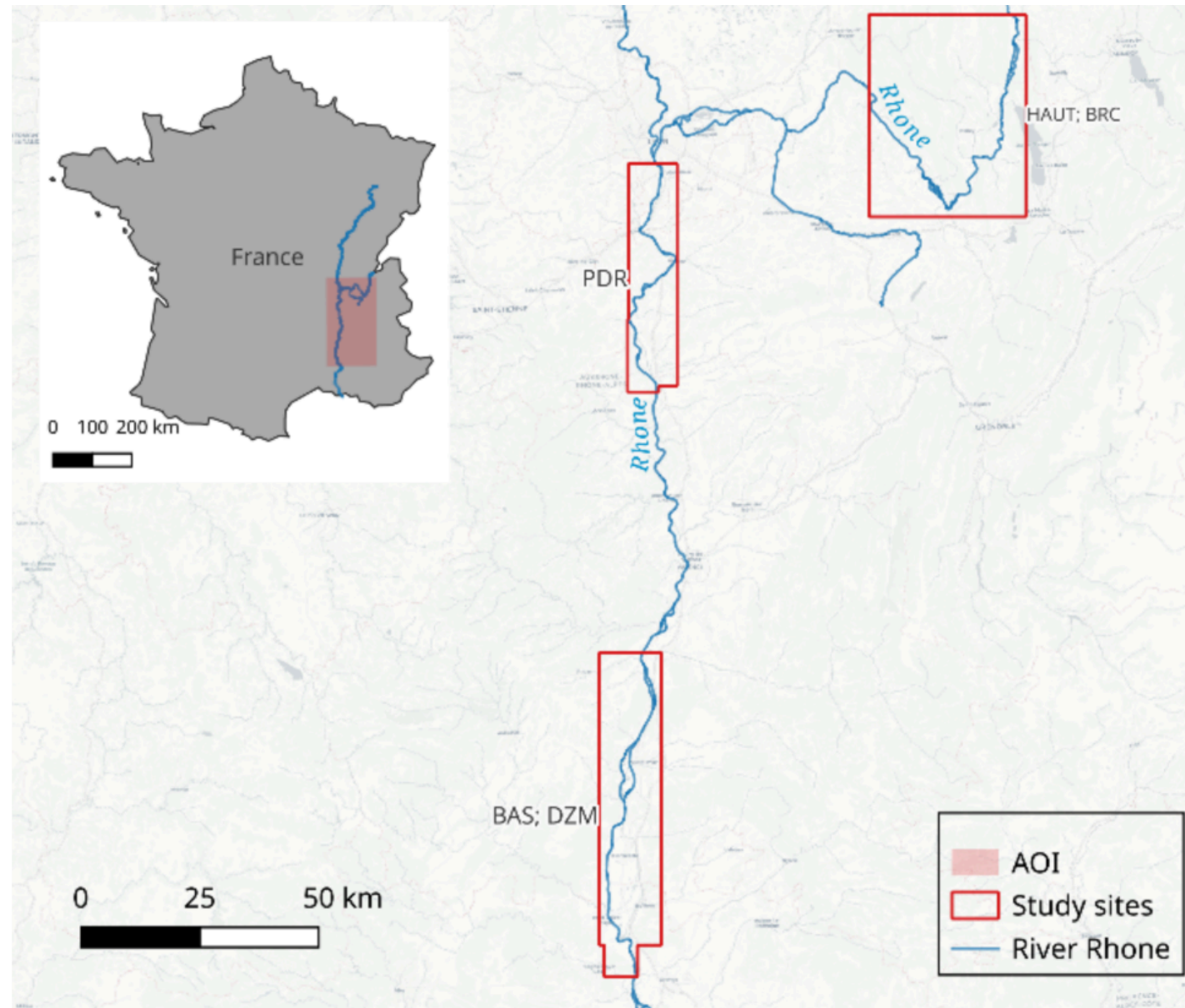
- Satellite imagery: coarse-resolution
- Continuous large-scale observation

How do we bridge this gap?

Objectives

- **Benchmark** state-of-the-art SISR models on real-world TIR data
- Integrate **auxiliary spatial and temporal** information to **guide** high-quality reconstruction
- Evaluate **recurrent super-resolution** networks for physically consistent reconstruction

Study Area & Data

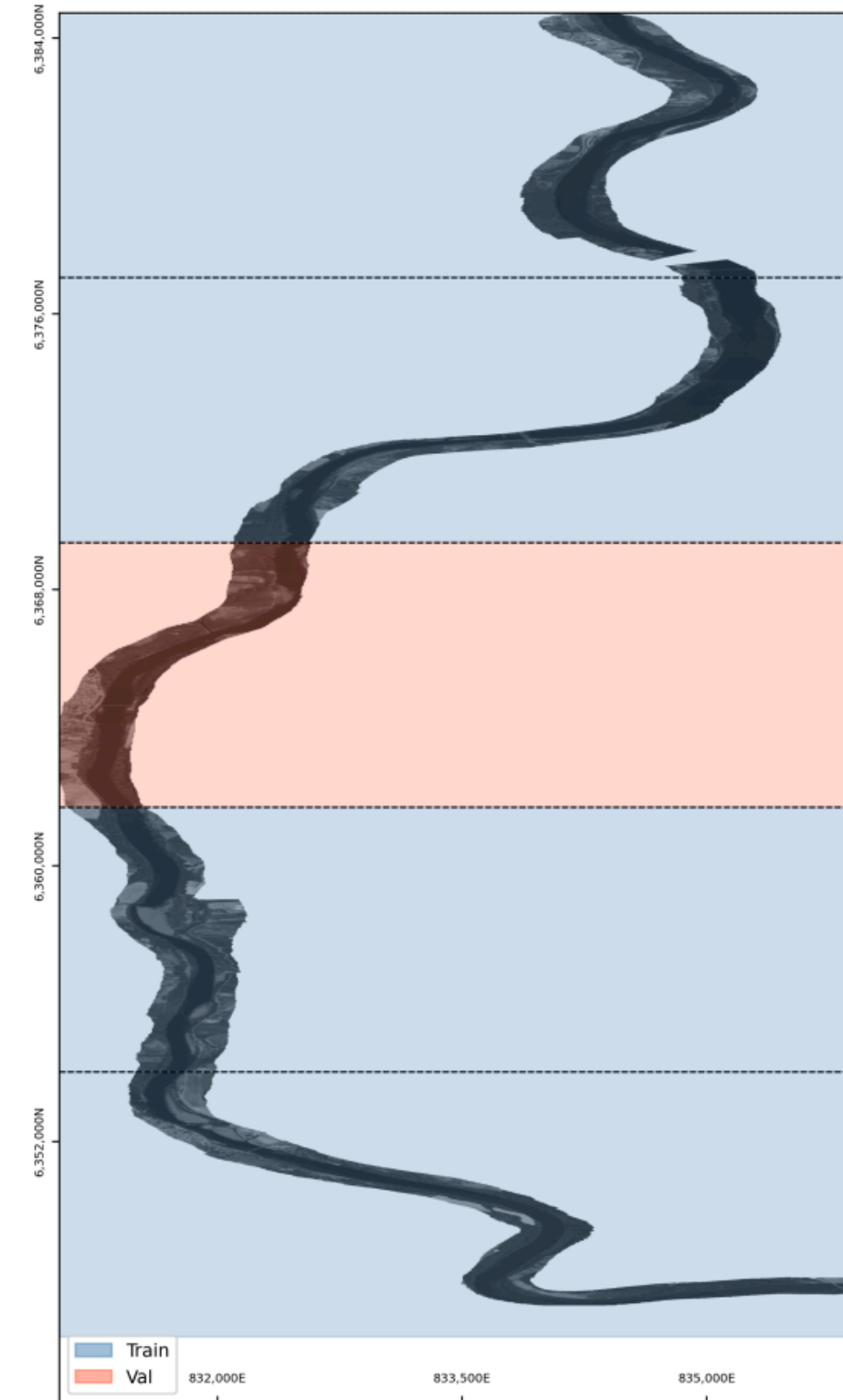


- Rhone River, France : 11 airborne TIR campaigns
- **HR reference** : Airborne TIR at 0.36-1.60m resolution
- **LR input**: Landsat 8/9 TIR at 30m resolution
- **Auxiliary data**: Optical bands, indices, land cover, DEM

Temporal mismatch: Real-world acquisition time differences

Data

- Super-resolution task:
 - scale factor: x4
 - Target resolution: 30m (LR) → 7.5m (HR)
- Data Split
 - Geographic holdout (Test): BRC + HAUT
 - Spatial block split : 80% Train/ 20% val
- Patches
 - HR - 256 x 256
 - LR - 64x64



Spatial blocks of train(blue) and val(red)

Benchmarking SISR Methods

Convolutional (CNN)	Transformer-based	Generative (GAN)
EDSR	SwinIR, HAT	Real-ESRGAN
Local receptive fields; fast baseline	Long-range dependencies; self- attention and cross-attention	Adversarial training, perceptual realism

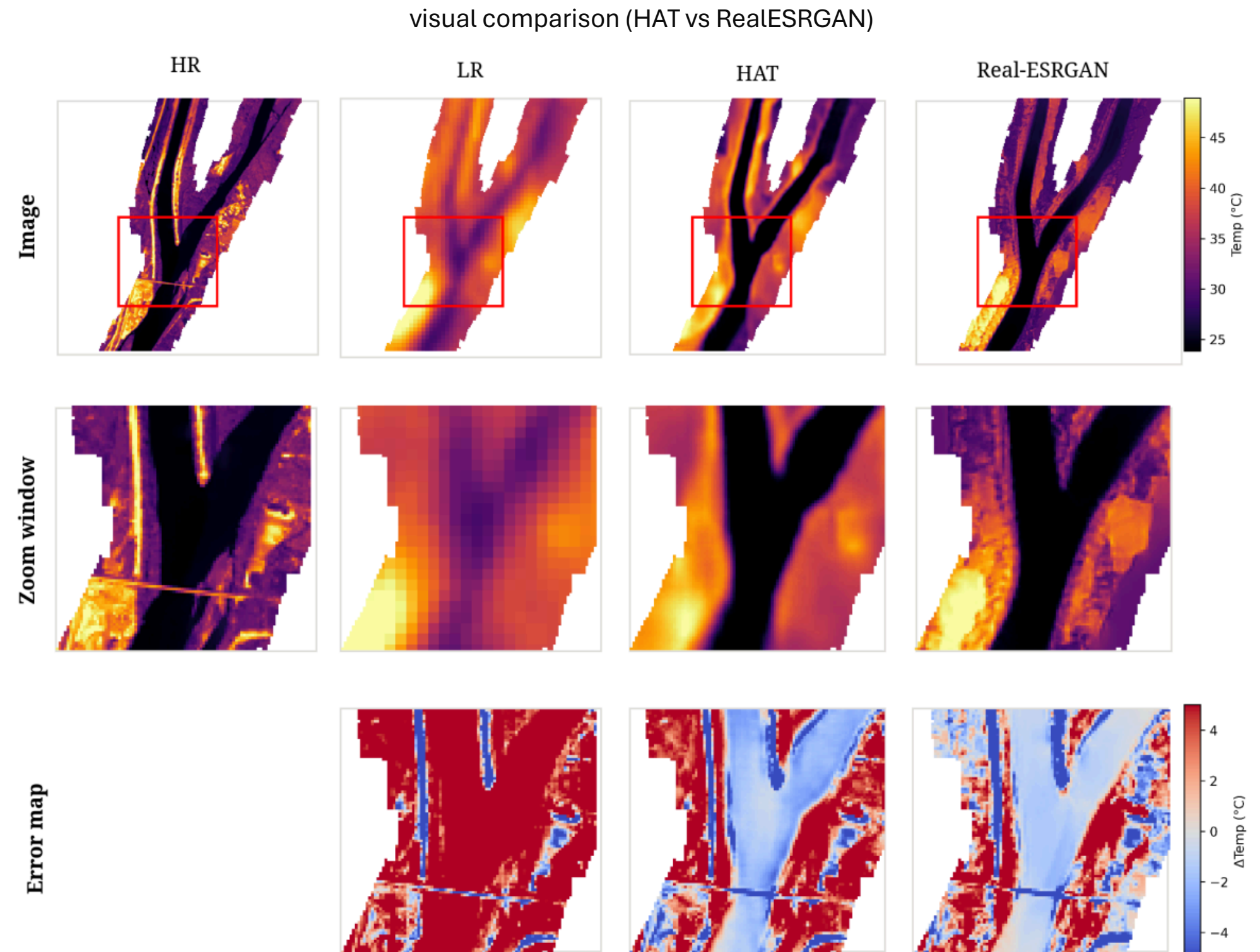
Perception-distortion tradeoff¹: pixel accuracy vs perceptual realism

1. Y. Blau, R. Mechrez, R. Timofte, T. Michaeli, and L. Zelnik-Manor, "The 2018 PIRM Challenge on Perceptual Image Super-Resolution," in Computer Vision – ECCV 2018 Workshops, L. Leal-Taixé and S. Roth, Eds., Cham: Springer International Publishing, 2019, pp. 334–355, isbn: 978-3-030-11021-5. doi: 10.1007/978-3-030-11021-5_21.

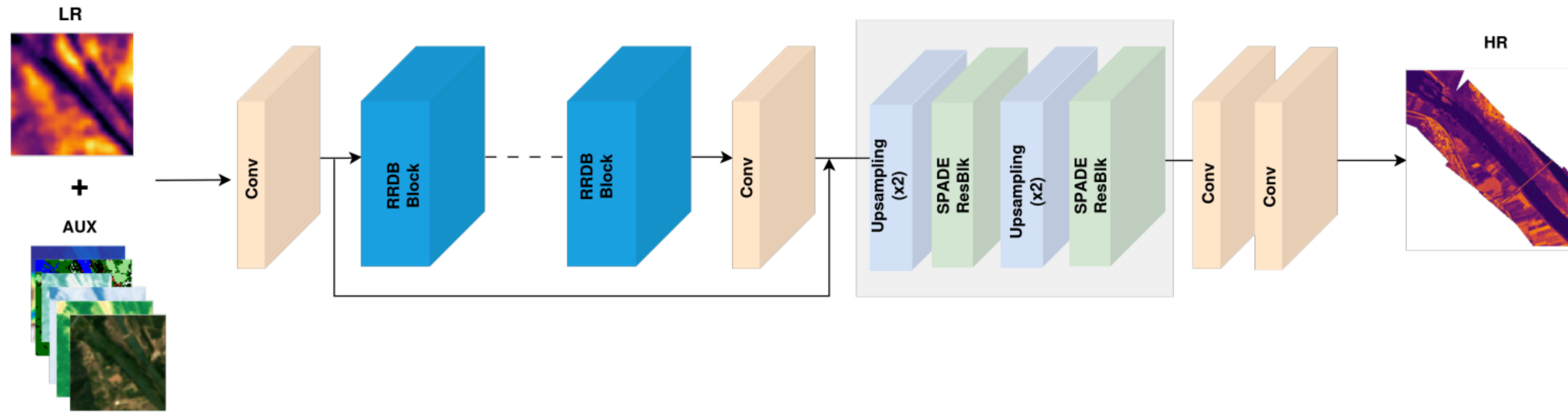
Benchmarking Results

Model	Water Region				
	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	LPIPS ↓
EDSR	2.54	3.50	18.41	0.520	0.609
SwinIR	2.27	3.44	18.56	0.546	0.552
HAT	2.34	3.38	18.96	0.541	0.553
Real-ESRGAN	2.52	3.57	18.17	0.302	0.402

- Transformer: best pixel-level metrics
- GAN: best perceptual quality
- Baseline choice: Real-ESRGAN - preserves high-frequency details



Guided Super-Resolution



Guided Real-ESRGAN¹ (generator)

Early Fusion: Auxiliary data fused with LR TIR at input

SPADE² conditioning: injects high-resolution land cover to guide high quality reconstruction.

U-Net discriminator¹: provides per-pixel ‘realness’ feedback for local detail enhancement.

1. X. Wang, L. Xie, C. Dong, and Y. Shan, “Real-ESRGAN: Training Real-World Blind Super-Resolution with Pure Synthetic Data,” in 2021 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), Oct. 2021, pp. 1905–1914. doi: 10.1109/ICCVW54120.2021.00217.

2. T. Park, M.-Y. Liu, T.-C. Wang, and J.-Y. Zhu, “Semantic Image Synthesis With Spatially-Adaptive Normalization,” in 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun. 2019, pp. 2332–2341. doi: 10.1109/CVPR.2019.00244.

Loss Formulation

$$\mathcal{L}_{total} = \mathcal{L}_{pixel} + \lambda_{perc}\mathcal{L}_{perc} + \lambda_{adv}\mathcal{L}_{adv}$$

- Multi-term loss: Combines pixel-level reconstruction with perceptual and adversarial terms

$$\mathcal{L}_{pixel} = w_t \cdot (\mathcal{L}_{rec} + \lambda_{grad}\mathcal{L}_{grad})$$

$$\mathcal{L}_{rec} = \lambda_w\mathcal{L}_w + \lambda_{nw}\mathcal{L}_{nw}$$

- Region decoupling : Separates water and land to prioritize water-surface reconstruction.
- Gradient Loss to preserve sharp thermal transitions/edges.

$$t_{\Delta} = \lambda_a \hat{t}_{\Delta}^{DOY} + \lambda_b \hat{t}_{\Delta}^{TOD}, \quad w_t = 1 - t_{\Delta}$$

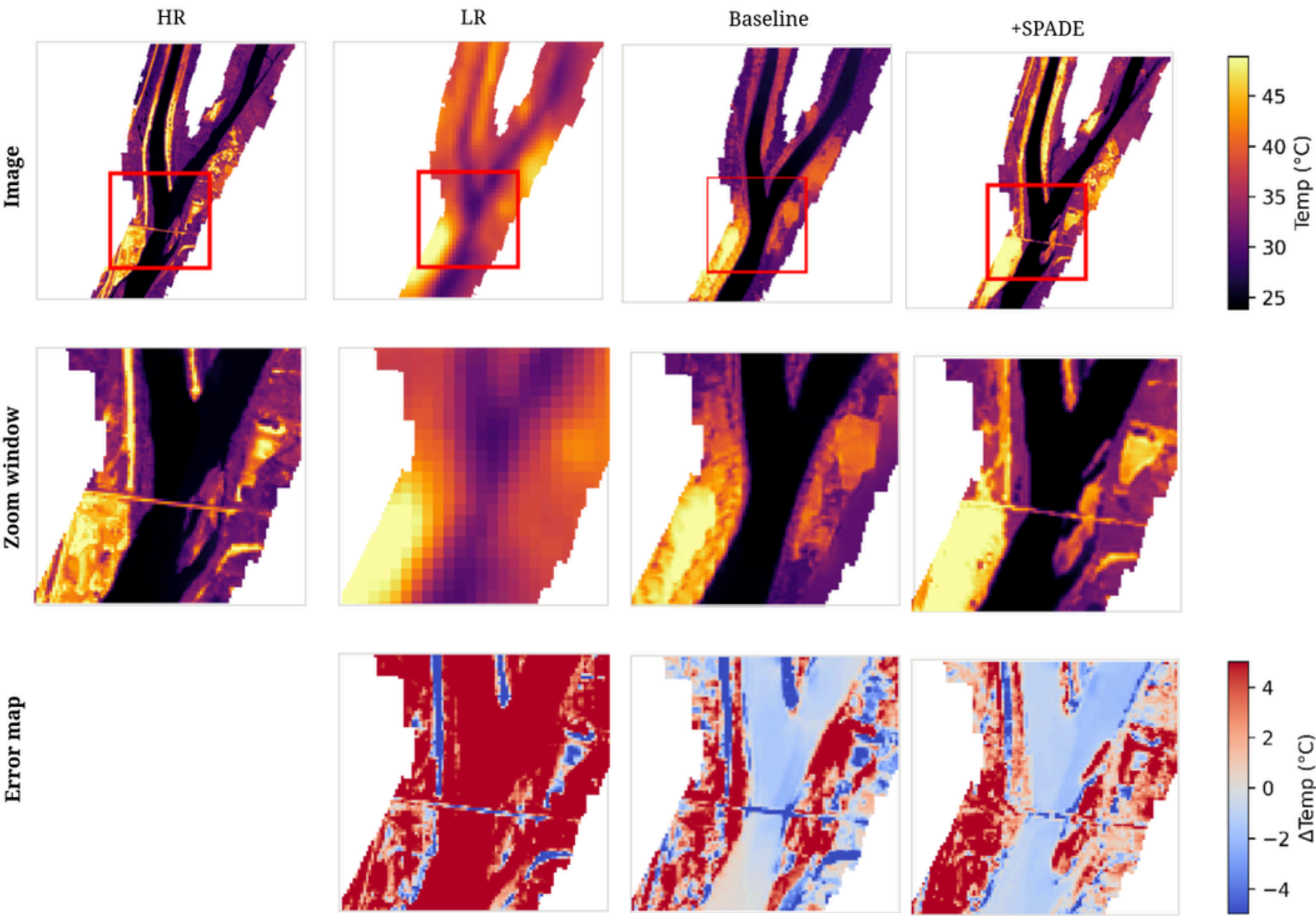
- Difference can be real temperature change not only resolution.
- Temporal weighting: Penalizes thermal gaps based on time-of-day and day-of-year.

Guided SR Results

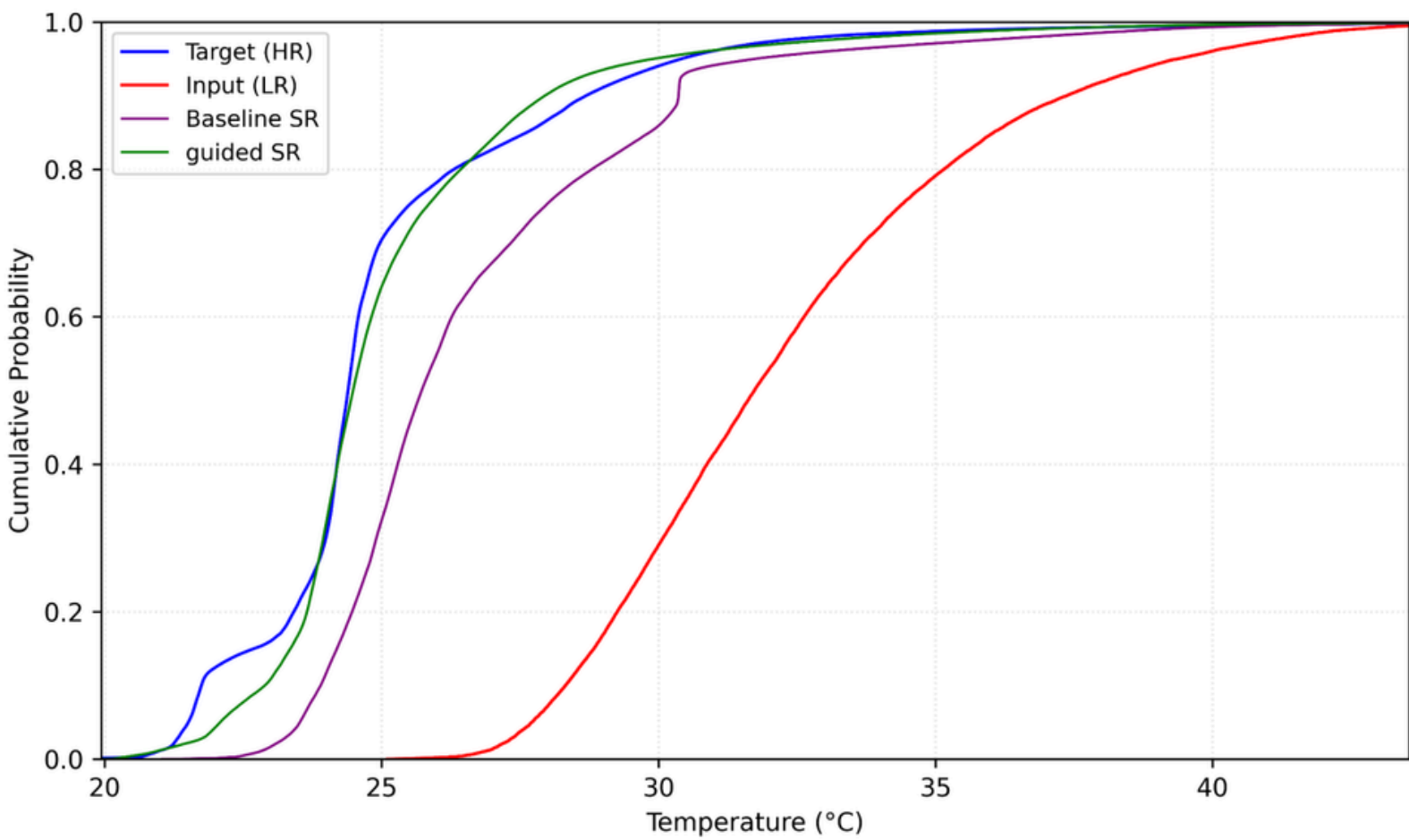
Water Region

Model	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Real-ESRGAN	2.52	3.57	18.17	0.302	0.402
+SPADE	1.74	2.45	21.52	0.511	0.461

visual comparison (baseline vs guided Real-ESRGAN)



Temperature distribution of water region

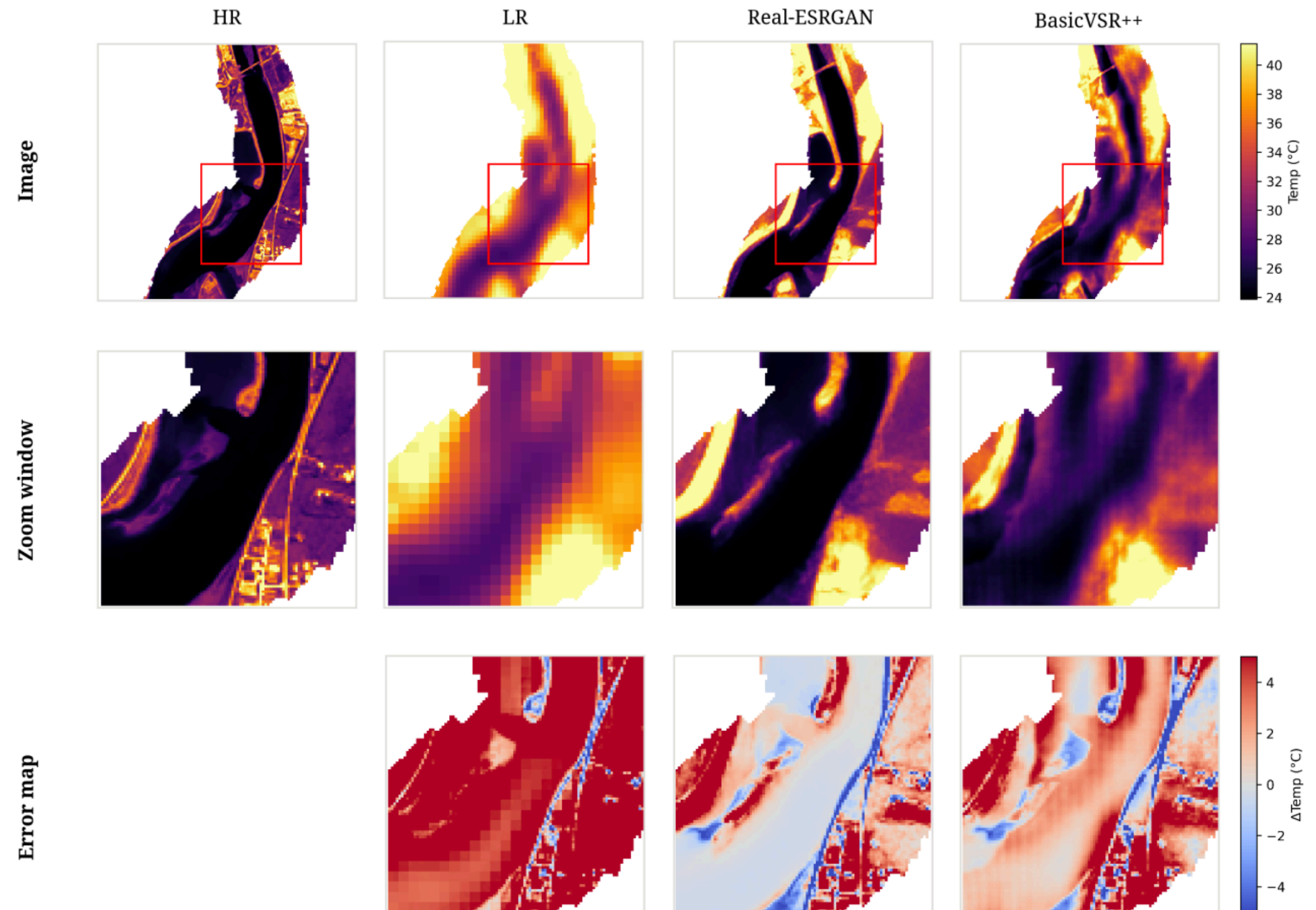


Sequential SR

- Does neighbouring spatial context along river profile improve thermal reconstruction?
- Adapted BasicVSR++¹, a bi-directional video SR framework, for sequential TIR tiles.

Model	Water Region				
	MAE ↓	RMSE ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Real-ESRGAN	1.74	2.45	21.52	0.511	0.461
BasicVSR++	1.97	2.65	20.84	0.477	0.505

visual comparison (guided Real-ESRGAN and guided BasicVSR++)

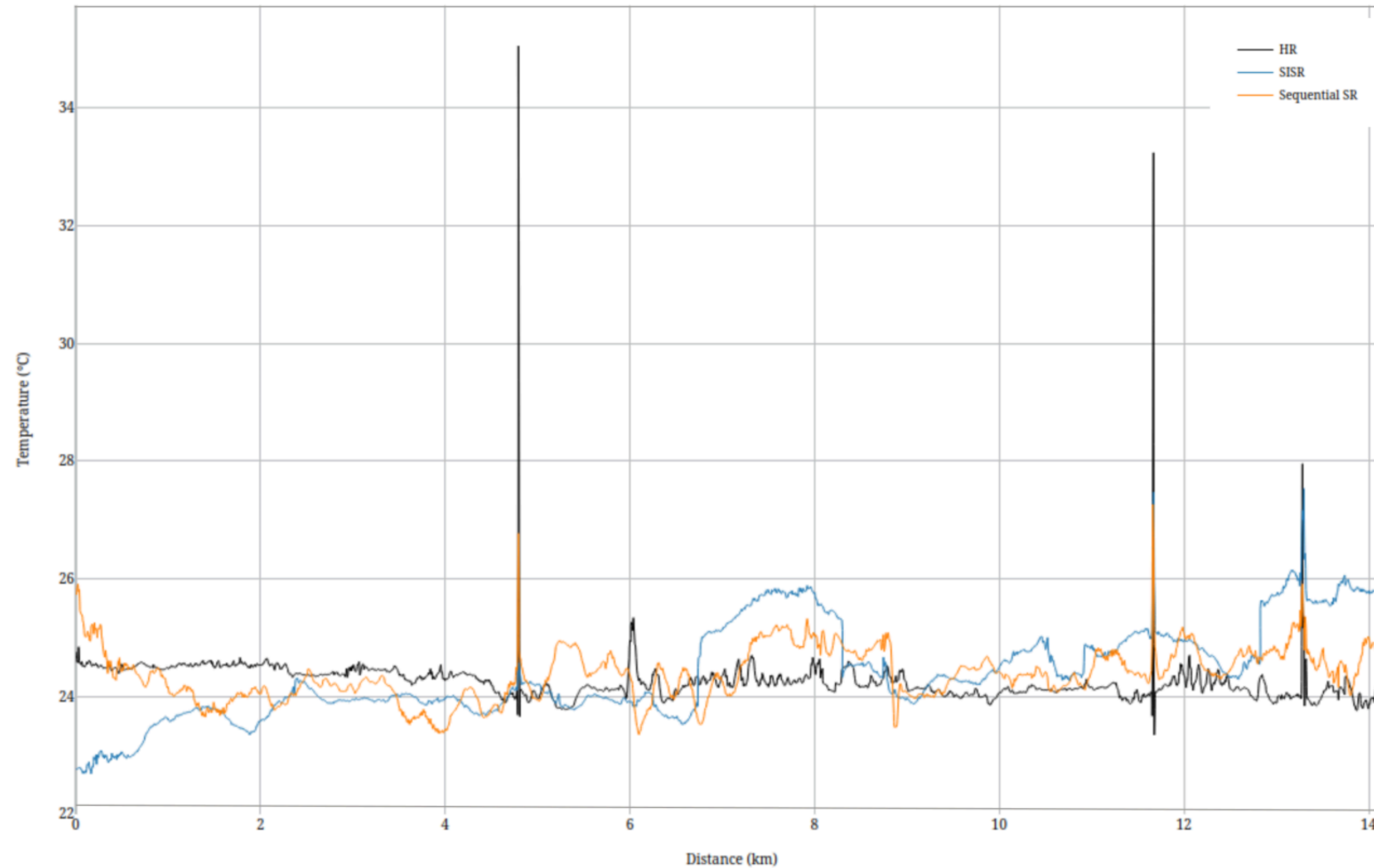


1. K. C. Chan, S. Zhou, X. Xu, and C. C. Loy, "BasicVSR++: Improving Video Super-Resolution with Enhanced Propagation and Alignment," in 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun. 2022, pp. 5962–5971.
doi: 10.1109/CVPR52688.2022.00588.

Sequential SR

- Closely reconstructs spatial continuity of thermal patterns along the river profile

Temperature profile along 14km river reach (BRC_2022 campaign)



Key Insights

- Standard SR models require specific adaptation to complex real-world TIR domain.
- Guided Real-ESRGAN significantly improves thermal infrared reconstruction.
- Perception-distortion tradeoff: Generative model require careful tuning.
- Sequential SR potential to produced thermal profile closely matching HR reference.

Limitations

- Temporal Mismatch between image pairs
- Performance variation across held-out test
- Video-based SR introduces temporal assumptions.

Future Work

- Training on larger, diverse thermal regimes and rivers.
- Sequential SR tailored to spatial river sequences.

Thank You!

Question and Discussion



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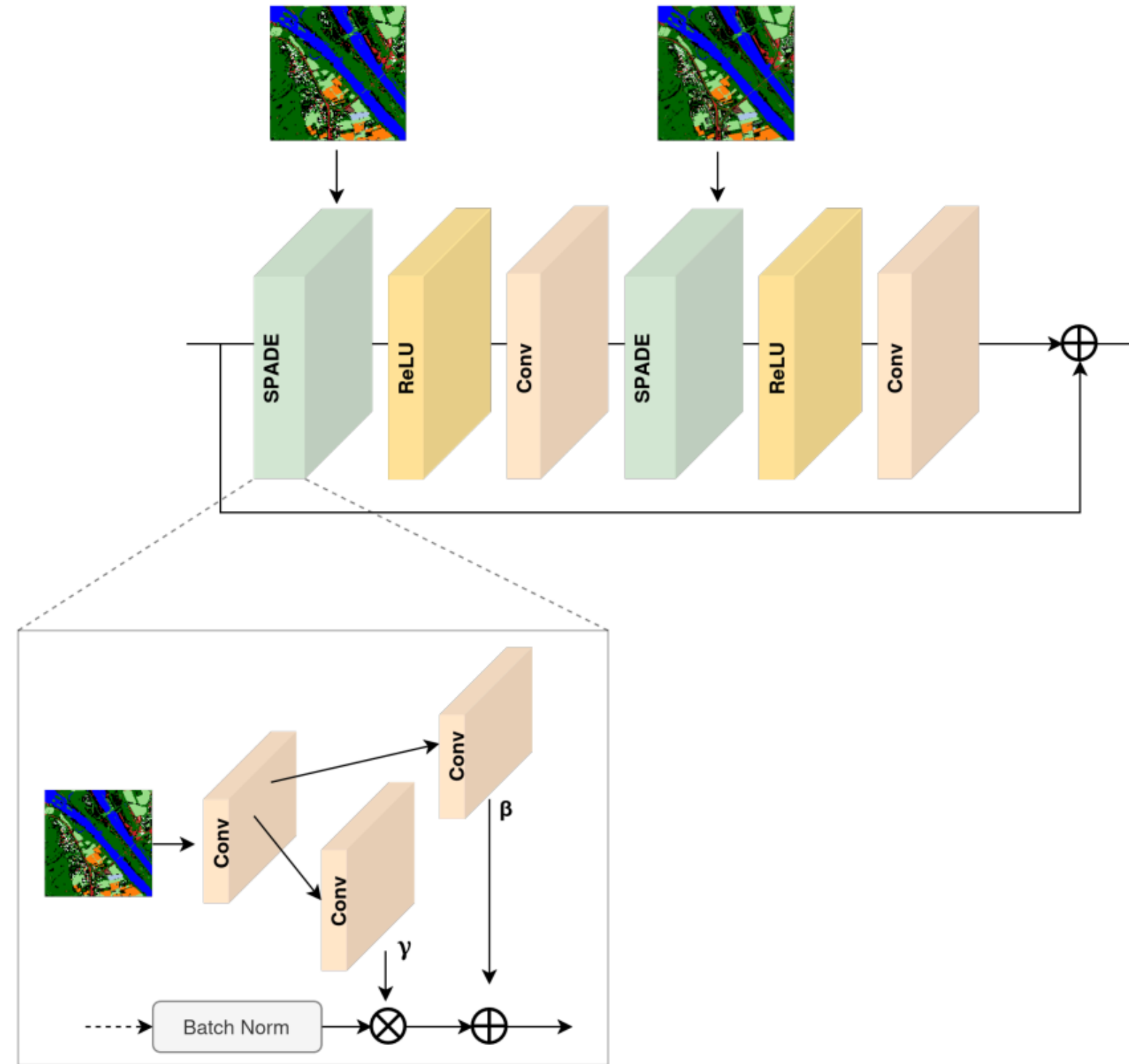
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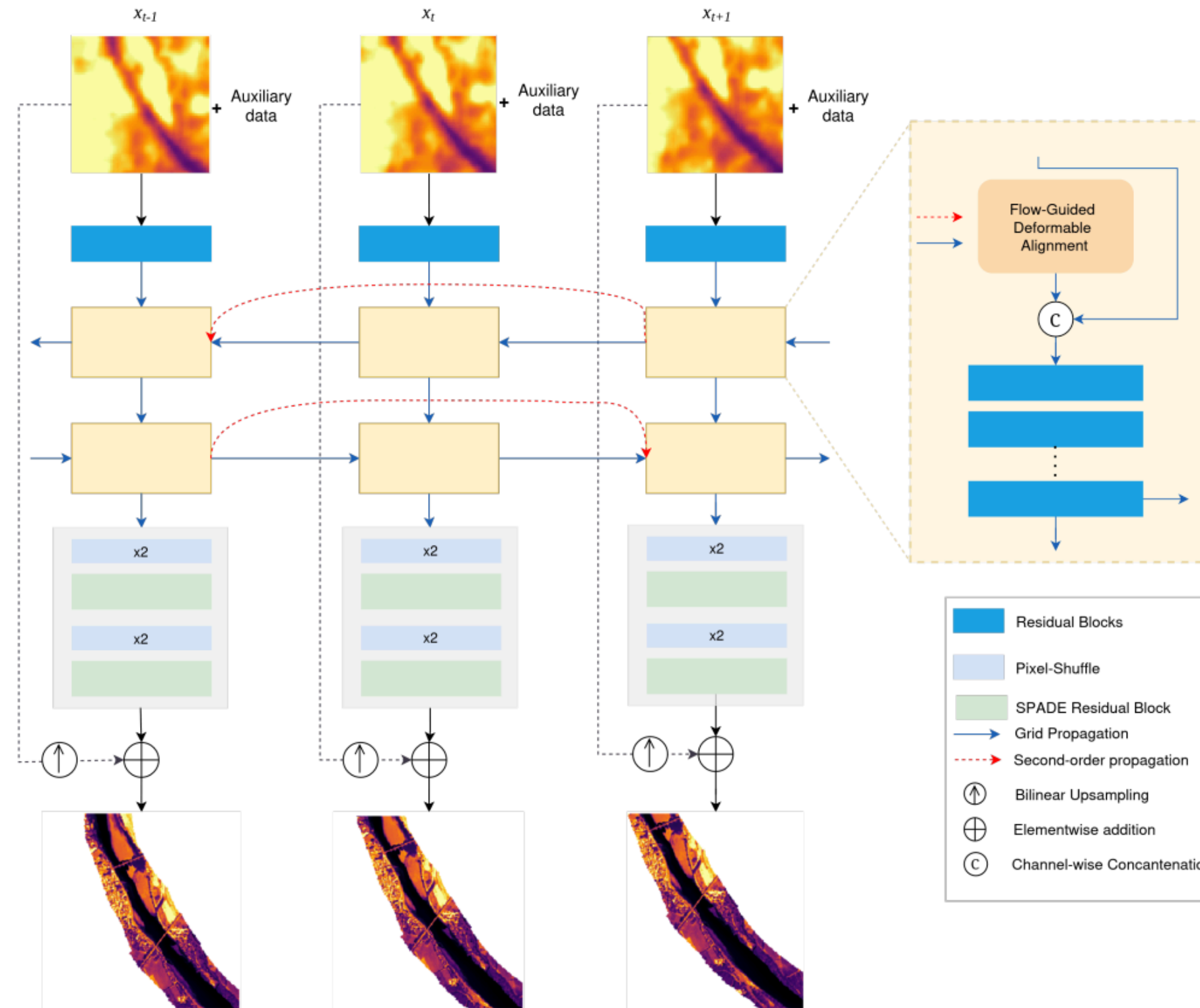


SPADE



SPADE Residual block structure

Sequential SR



Simplified guided BasicVSR++ architecture